Supervised Learning: Classification

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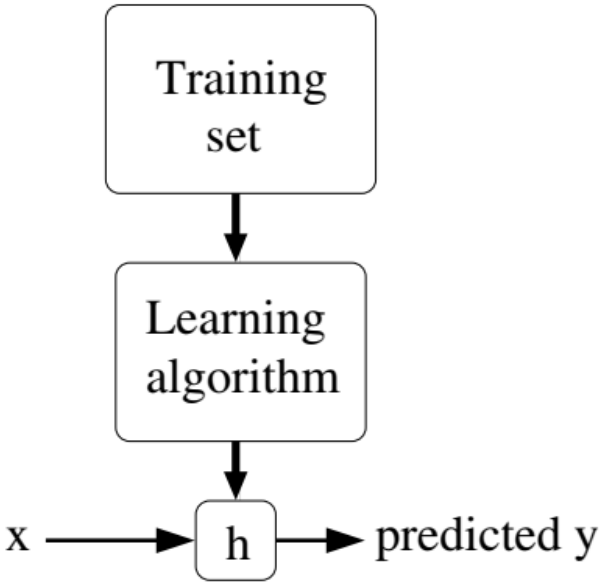
Outline

- Apa itu Supervised Learning?
- Task dalam Supervised Learning
- Klasifikasi
- Logistic Regression

Apa itu Supervised Learning?

- Proses pembelajaran mesin yang mencoba membuat fungsi pemetaan dari input menjadi output berdasarkan data training



Task dalam Supervised Learning

- Regression → memprediksi nilai continue, seperti harga rumah, harga saham, dll

- Classfication → Memprediksi nilai diskrit, seperti kategori, kelas, kelompok, dll

Apakah ini foto Susi, Steak, atau Pizza?

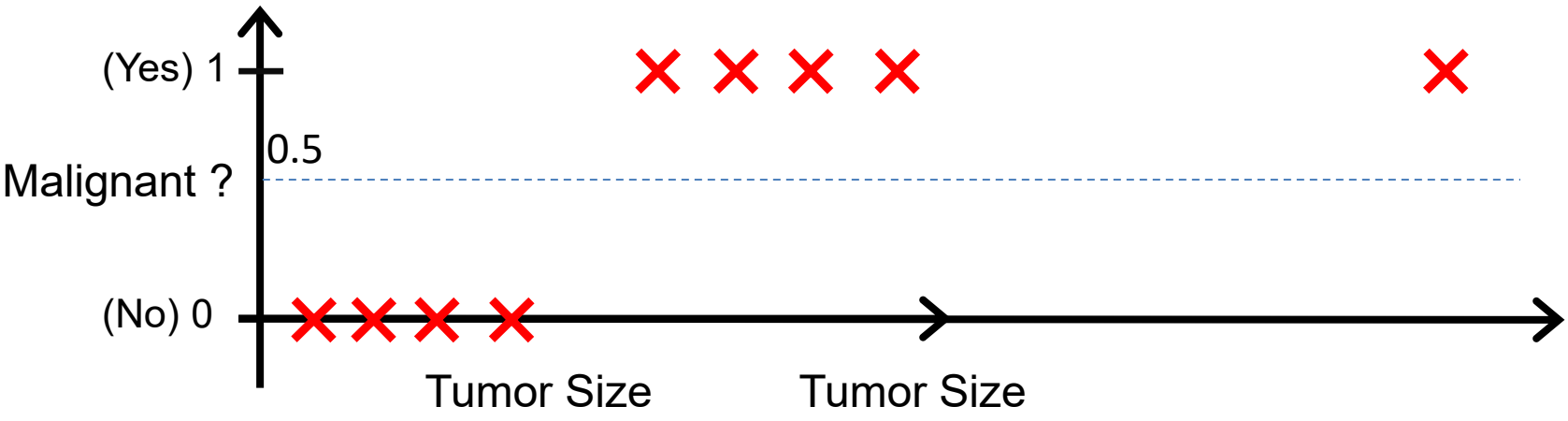
Klasifikasi

- Email: Spam / Bukan Spam?
- Online Transactions: Fraudulent (Yes / No)?
- Tumor: Malignant / Benign ?

$y \in \{0, 1\}$

0: “Negative Class” (e.g., benign tumor)
1: “Positive Class” (e.g., malignant tumor)

Klasifikasi: Tumor



Threshold output classifier $h_w(x)$ pada 0.5:

If $h_w(x) \geq 0.5$, predict "y = 1"

If $h_w(x) < 0.5$, predict "y = 0"

Problem Formulation

Classification: y = 0 or 1

$h_w(x)$ can be > 1 or < 0

Bagaimana Interpretasinya?

Logistic Regression: $0 \leq h_w(x) \leq 1$

Bisa diinterpretasikan sbg probabilitas

Logistic Regression

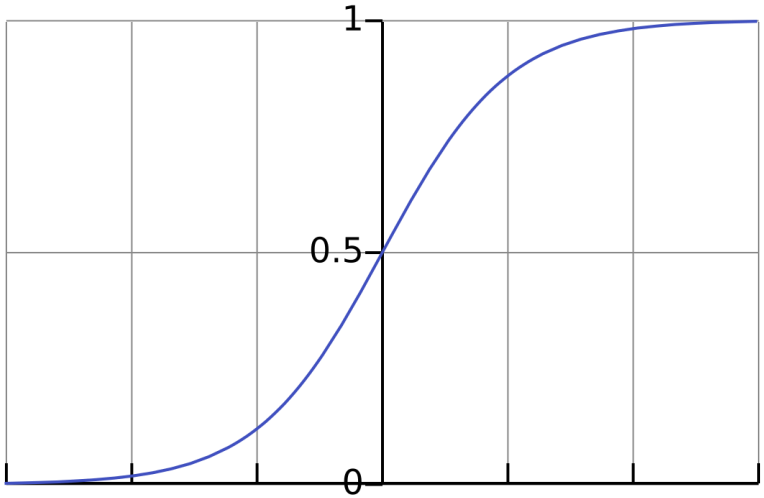
Want $0 \leq h_w(x) \leq 1$

$$g(z) = \frac{1}{1 + e^{-z}}$$

$$h_w(x) = w^T x$$

$$h_w(x) = g(w^T x)$$

Sigmoid function
Logistic function



Logistic Regression

$h_w(x)$ = probabilitas y = 1 yang diestimasi berdasarkan input x

Contoh: If $x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \text{tumorSize} \end{bmatrix}$

$$h_w(x) = 0.7$$

Artinya probabilitas pasien terkena Malignant adalah 70%

"Probailitas y=1, given x, dengan parameter w" → $P(y = 1|x; w)$

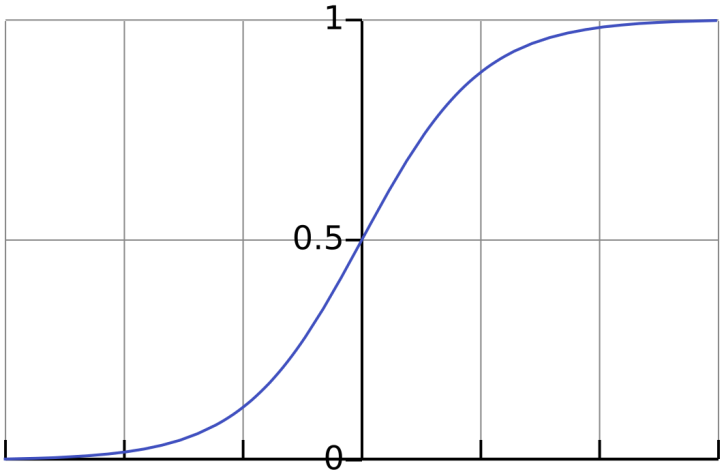
$$P(y = 0|x; w) + P(y = 1|x; w) = 1$$

$$P(y = 0|x; w) = 1 - P(y = 1|x; w)$$

Logistic Regression

$$h_w(x) = g(w^T x)$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

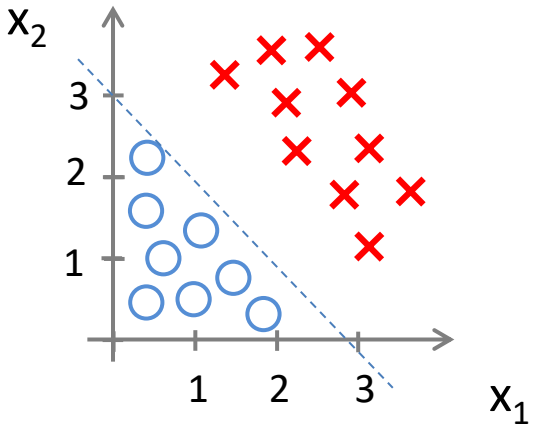


Maka:

predict "$y = 1$" jika $h_w(x) \geq 0.5$

predict "$y = 0$" jika $h_w(x) < 0.5$

Decision Boundary



$$h_w(x) = g(w_0 + w_1 x_1 + w_2 x_2)$$

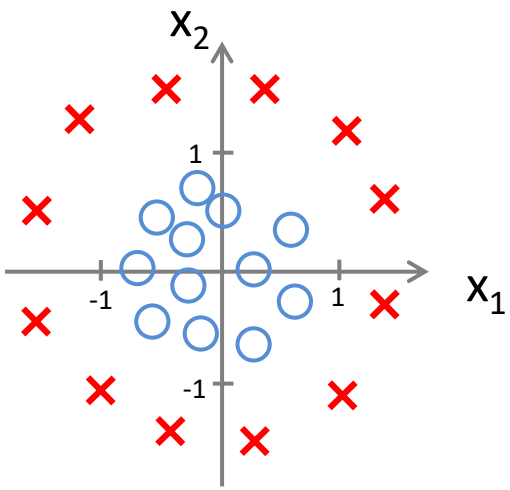
$$w_0 = -3$$

$$w_1 = 1$$

$$w_2 = 1$$

Predict "$y = 1$" if $-3 + x_1 + x_2 \geq 0$

Non-linear decision boundaries



Cost Function

Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \cdots, (x^{(m)}, y^{(m)})\}$

m examples $x \in \begin{bmatrix} x_0 \\ x_1 \\ \cdots \\ x_n \end{bmatrix}$ $x_0 = 1, y \in \{0, 1\}$

$$h_w(x) = g(w^T x)$$

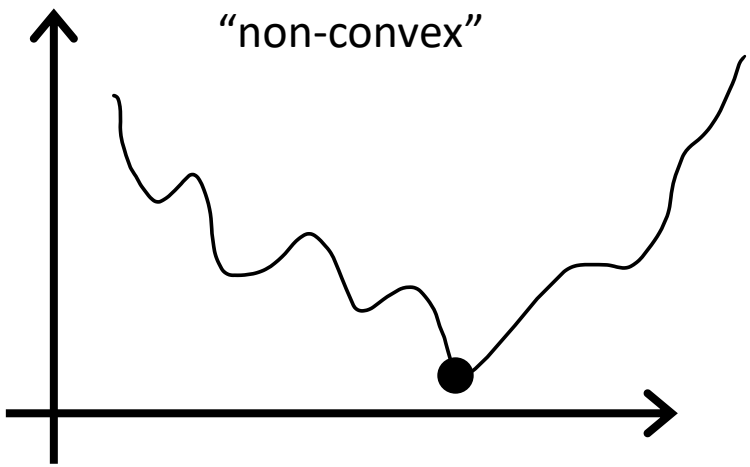
$$g(z) = \frac{1}{1 + e^{-z}}$$

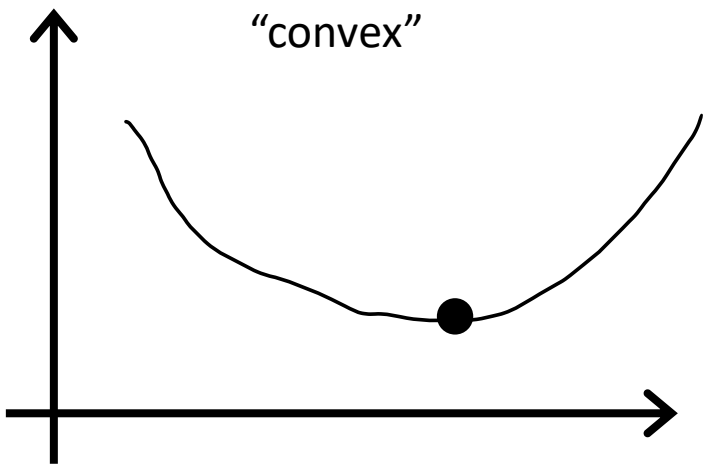
Bagaimana memilih w yang tepat?

Cost Function

Linear regression: $$J(w_0, w_1) = \frac{1}{2m}\sum_{i=1}^{m}(h_w(x^{(i)}) - y^{(i)})^2$$

Cost: $$J(w) = \frac{1}{2}(h_w(x^{(i)}) - y^{(i)})^2$$





Logistic Regression Cost Function

$$Cost(h_w(x^{(i)}), y^{(i)}) = \begin{cases} -\log\left(h_w(x^{(i)})\right) & if\ y = 1 \\ -\log\left(1 - h_w(x^{(i)})\right) & if\ y = 0 \end{cases}$$

$$J(w) = \frac{1}{m}\sum_{i=1}^{m} Cost(h_w(x^{(i)}), y^{(i)})$$

$$h_w(x) = g(w^T x)$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

$$J(w) = -\frac{1}{m}\left[\sum_{i=1}^{m} y^{(i)} \log h_w(x^{(i)}) + (1 - y^{(i)})\log(1 - h_w(x^{(i)}))\right]$$

Gradient dari Loss Function

$$J(w) = -\frac{1}{m}\left[\sum_{i=1}^{m} y^{(i)} \log h_w\left(x^{(i)}\right) + \left(1 - y^{(i)}\right)\log\left(1 - h_w\left(x^{(i)}\right)\right)\right]$$

dimana $h_w(x) = g(w^T x)$ dan $g(z) = \dfrac{1}{1 + e^{-z}}$

Misal $\boldsymbol{a} = \boldsymbol{h_w}\left(\boldsymbol{x}^{(i)}\right) = \boldsymbol{g}(\boldsymbol{z})$, $\boldsymbol{z} = \boldsymbol{w}^T\boldsymbol{x}$, $\mathbf{b} = \boldsymbol{y}^{(i)}$

$$J(w) = \frac{1}{m}\left[\sum_{i=1}^{m} -\{b \ \log a + (1 - b)\log(1 - a)\}\right]$$

$$\frac{\partial J(w)}{\partial w_j} = \frac{\partial}{\partial w_j}\left\{\frac{1}{m}\left[\sum_{i=1}^{m} -\{b \ \log a + (1 - b)\log(1 - a)\}\right]\right\}$$

$$\frac{\partial J(w)}{\partial w_j} = \frac{1}{m}\sum_{i=1}^{m} \frac{\partial}{\partial w_j}\{-(b \ \log a + (1 - b)\log(1 - a))\}$$

Gradient dari Loss Function

$$L = -[b\ \log a + (1-b)\log(1-a)]$$

$$\boldsymbol{a = h_w\left(x^{(i)}\right) = g(z)}\ ,\quad \boldsymbol{g(z) = \frac{1}{1+e^{-z}}}\ ,\quad \boldsymbol{z = w^T x},\quad \mathbf{b} = \boldsymbol{y^{(i)}}$$

$$\frac{\partial J(w)}{\partial w_j} = \frac{1}{m}\sum_{i=1}^{m}\frac{\partial}{\partial w_j}L$$

$$\frac{\partial L}{\partial w_j} = \frac{\partial L}{\partial a}\cdot\frac{\partial a}{\partial w_j} = \frac{\partial L}{\partial a}\cdot\frac{\partial a}{\partial z}\cdot\frac{\partial z}{\partial w_j}$$

$$\frac{\partial L}{\partial a} = -\frac{b}{a} - \left(\frac{1-b}{1-a}\cdot -1\right) = -\frac{b}{a} + \frac{1-b}{1-a}$$

$$y = f(g(x))$$

$$y = f(u),\quad u = g(x)$$

$$\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx}$$

$$\frac{\partial x^n}{\partial x} = n\,x^{n-1}$$

$$\frac{\partial \frac{1}{x}}{\partial x} = \frac{\partial x^{-1}}{\partial x} = -\frac{1}{x^2}$$

$$\frac{\partial \log(k.x)}{\partial x} = \frac{k}{x}$$

$$\frac{\partial\, e^x}{\partial x} = e^x$$

$$\frac{\partial\, e^{-x}}{\partial x} = -e^{-x}$$

$$\frac{\partial\,[f(x)]^n}{\partial x} = n\,[f(x)]^{n-1}\cdot\frac{\partial f(x)}{\partial x}$$

Gradient dari Loss Function

$$L = -[b\ \log a + (1-b)\log(1-a)]$$

$$\boldsymbol{a = h_w\left(x^{(i)}\right) = g(z)}\ ,\quad \boldsymbol{g(z) = \frac{1}{1+e^{-z}}}\ ,\quad \boldsymbol{z = w^T x},\quad \mathbf{b} = \boldsymbol{y^{(i)}}$$

$$\frac{\partial L}{\partial w_j} = \frac{\partial L}{\partial a}\cdot\frac{\partial a}{\partial w_j} = \frac{\partial L}{\partial a}\cdot\frac{\partial a}{\partial z}\cdot\frac{\partial z}{\partial w_j}$$

$$\frac{\partial a}{\partial z} = \frac{\partial}{\partial z}(1+e^{-z})^{-1} = -1.(1+e^{-z})^{-2}\ .\ \frac{\partial}{\partial z}(1+e^{-z})$$

$$= -(1+e^{-z})^{-2}\ .\ -e^{-z}$$

$$= \frac{1}{(1+e^{-z})^2}\ .e^{-z}$$

$$= a^2.e^{-z}$$

$$= a^2.\left(\frac{1-a}{a}\right)$$

$$= a\,(1-a)$$

$$a = g(z) = \frac{1}{1+e^{-z}}$$

$$a^2 = \frac{1}{(1+e^{-z})^2}$$

$$a(1+e^{-z}) = 1$$

$$a + ae^{-z} = 1$$

$$ae^{-z} = 1-a$$

$$e^{-z} = \frac{1-a}{a}$$

$$\frac{\partial x^n}{\partial x} = n\,x^{n-1}$$

$$\frac{\partial \frac{1}{x}}{\partial x} = \frac{\partial x^{-1}}{\partial x} = -\frac{1}{x^2}$$

$$\frac{\partial \log(k.x)}{\partial x} = \frac{k}{x}$$

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$$\frac{\partial\,[f(x)]^n}{\partial x} = n\,[f(x)]^{n-1}\,.\frac{\partial f(x)}{\partial x}$$

Gradient dari Loss Function

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$$\frac{\partial L}{\partial w_j} = \frac{\partial L}{\partial a}\cdot\frac{\partial a}{\partial w_j} = \frac{\partial L}{\partial a}\cdot\frac{\partial a}{\partial z}\cdot\frac{\partial z}{\partial w_j}$$

$$\frac{\partial z}{\partial w_j} = \frac{\partial}{\partial w_j} w^T x$$

$$\frac{\partial z}{\partial w_j} = \frac{\partial}{\partial w_j}\left(w_0 x_0^{(i)} + w_1 x_1^{(i)} + \ \dots\ + w_j x_j^{(i)}\right)$$

$$= x_j^{(i)}$$

Gradient dari Loss Function

$$L = -[b\ \log a + (1-b)\log(1-a)]$$

$$\boldsymbol{a = h_w\left(x^{(i)}\right) = g(z)}\ ,\quad \boldsymbol{g(z) = \frac{1}{1+e^{-z}}}\ ,\quad \boldsymbol{z = w^T x},\quad \mathbf{b} = \boldsymbol{y^{(i)}}$$

$$\frac{\partial L}{\partial w_j} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w_j}$$

$$= \left(-\frac{b}{a} + \frac{1-b}{1-a}\right) . \left(a\,(1-a)\right)\ .\ x_j^{(i)}$$

$$= \left(\frac{-b + ab + a\ - ab}{a(1-a)}\right) . a\,(1-a).\ x_j^{(i)}$$

$$= (-b + a).\ x_j^{(i)}$$

$$= (a - b).\ x_j^{(i)}$$

$$= \left(h_w(x^{(i)}) - y^{(i)}\right).\ x_j^{(i)}$$

Gradient dari Loss Function

$$J(w) = -\frac{1}{m}\left[\sum_{i=1}^{m} y^{(i)} \log h_w\left(x^{(i)}\right) + \left(1 - y^{(i)}\right)\log\left(1 - h_w\left(x^{(i)}\right)\right)\right]$$

dimana $h_w(x) = g(w^T x)$ dan $g(z) = \dfrac{1}{1 + e^{-z}}$

Misal $\boldsymbol{a} = \boldsymbol{h_w}\left(\boldsymbol{x}^{(i)}\right) = \boldsymbol{g}(\boldsymbol{z})$, $\boldsymbol{z} = \boldsymbol{w}^T\boldsymbol{x}$, $\mathbf{b} = \boldsymbol{y}^{(i)}$

$$L = -[b \ \log a + (1 - b)\log(1 - a)]$$

$$\frac{\partial J(w)}{\partial w_j} = \frac{1}{m}\sum_{i=1}^{m}\frac{\partial}{\partial w_j} L$$

$$\frac{\partial L}{\partial w_j} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w_j} = \left(h_w(x^{(i)}) - y^{(i)}\right).\ x_j^{(i)}$$

$$\frac{\partial J(w)}{\partial w_j} = \frac{1}{m}\sum_{i=1}^{m}\left(h_w(x^{(i)}) - y^{(i)}\right).\ x_j^{(i)}$$

Tipe Klasifikasi

Binary Classification

Spam atau Bukan?

from: cs@upi.edu

Hi, welcome to Deep Learning Course

from: bni@gmail.com

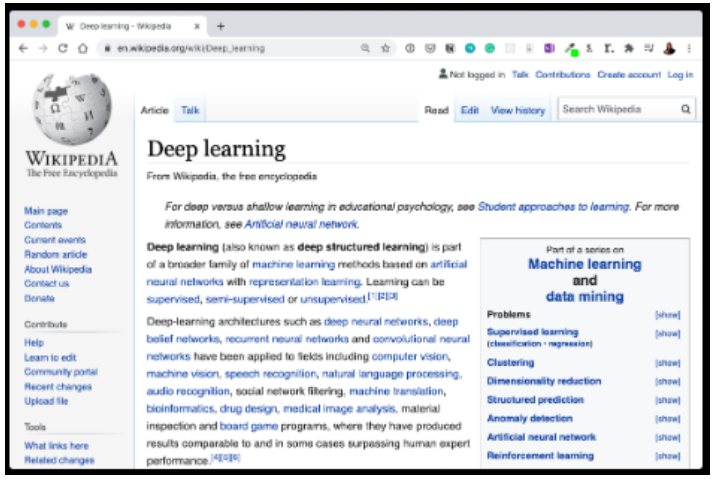
C0ongratu1ations! U win IDR 2.000.000

Multiclass Classification

Apakah ini foto Susi, Steak, atau Pizza?

Multilabel Classification

- Machine Learning
- Representation Learning
- Artificial Intelligence



Aneka Jenis Metric Evaluasi Klasifikasi

		True condition				
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$	
Predicted condition	Predicted condition positive	**True positive**, Power	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$	
	Predicted condition negative	**False negative**, Type II error	**True negative**	False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$	
		True positive rate (TPR), Recall, Sensitivity, probability of detection = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{TPR}{FPR}$	Diagnostic odds ratio (DOR) = $\frac{LR+}{LR-}$	F_1 score = $\frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$
		False negative rate (FNR), Miss rate = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) = $\frac{FNR}{TNR}$		

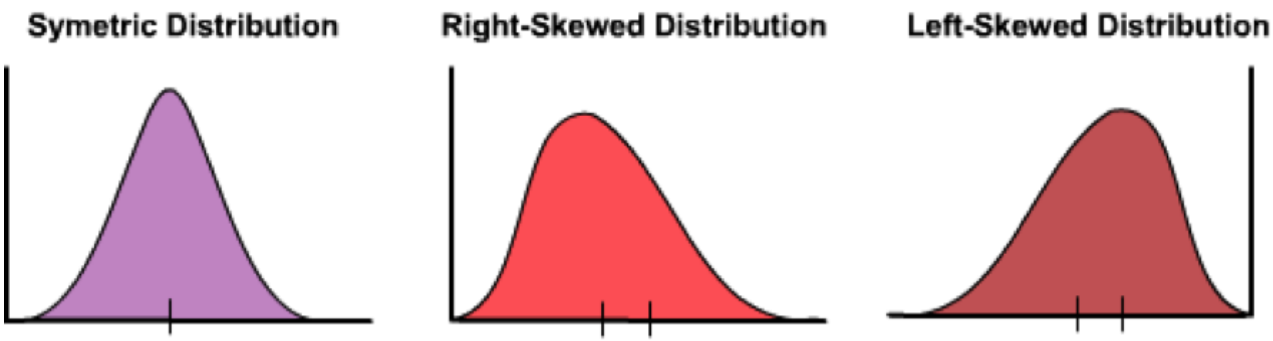
Confusion Matrix

		Actual	
		Positive	Negative
Predicted	Positive	**True Positive**	**False Positive**
	Negative	**False Negative**	**True Negative**

Accuracy

		Actual	
		Positive	Negative
Predicted	Positive	**True Positive**	**False Positive**
	Negative	**False Negative**	**True Negative**

- Accuracy = (TP+TN)/(TP+FP+FN+TN)
- Accuracy: Proporsi jml benar dibanding jml data
- Cocok utk data yg balance dan “not skewed”



Precision

		Actual	
		Positive	Negative
Predicted	Positive	**True Positive**	**False Positive**
	Negative	**False Negative**	**True Negative**

- Precision = (TP)/(TP+FP)
- Precision: Proporsi yg diprediksi positif dan benar-positif
- Digunakan jika kita ingin “sangat yakin” terhadap hasil prediksi kita
- Biasanya digunakan ketika harga kesalahan prediksi cukup mahal, sehingga kita perlu yakin betul dengan model yg dibuat

Recall

		Actual	
		Positive	Negative
Predicted	Positive	**True Positive**	**False Positive**
	Negative	**False Negative**	**True Negative**

- Recall = (TP)/(TP+FN)
- Recall: Proporsi kelas positif yang diklasifikan dg benar
- Cocok digunakan jika kita ingin meng-capture sebanyak mungkin kelas positif

F1-Score

$$F_1 = 2 * \frac{precision * recall}{precision + recall}$$

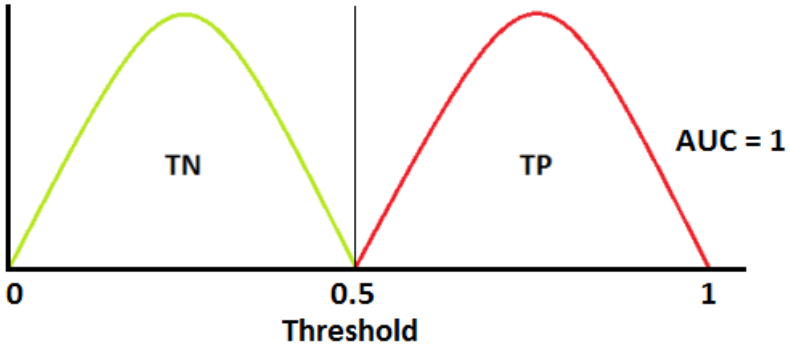
- Harmonic mean dari precision dan recall
- Memanfaatkan trade-off dari masing-masing metric: prec-rec
- Skornya berkisar antara 0 dan 1
- Cocok digunakan ketika kita menginginkan precision dan recall dari model kita baik

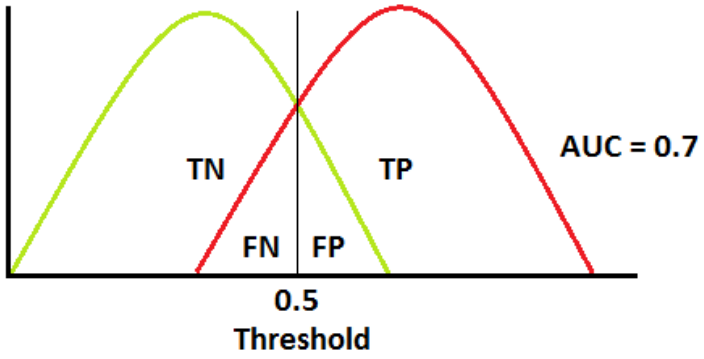
Weighted F1-Score

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{(\beta^2 \cdot \text{precision}) + \text{recall}}.$$

ROC

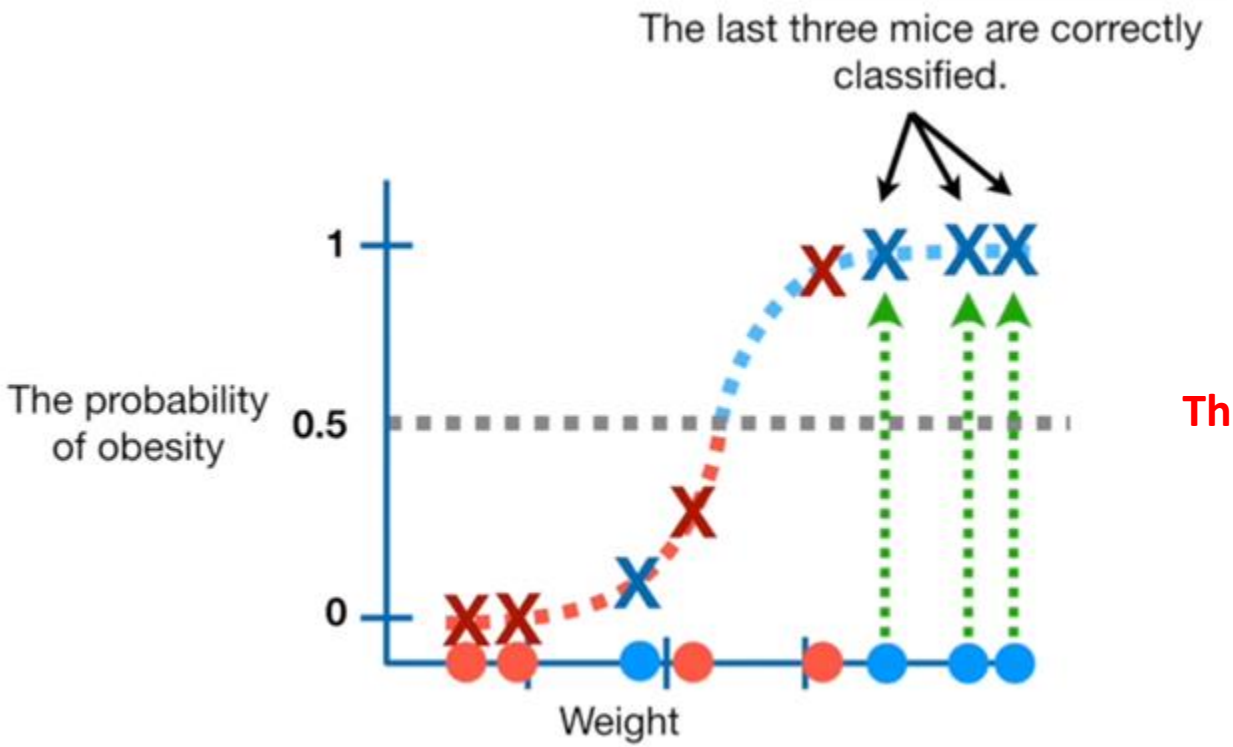
- ROC (Receiver Operating Characteristics): alat ukur performance untuk classification problem dalam menentukan threshold dari suatu model
- Default threshold: 0.5
- Scallable





ROC: Example

Kasus: Klasifikasi Tikus Obesitas/Tidak
Kelas: Biru → Obesitas
Merah → Tidak Obesitas



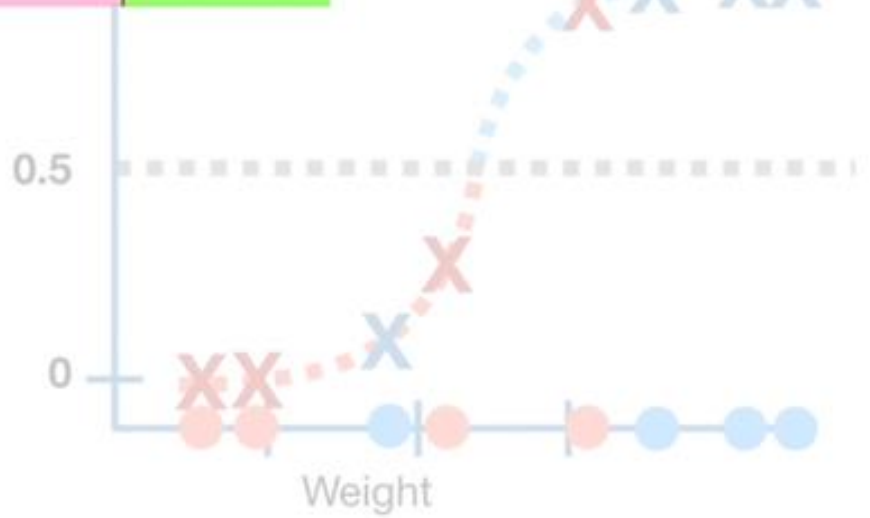
Threshold: 0.5

ROC: Example

Kasus: Klasifikasi Tikus Obesitas/Tidak
Kelas: Biru → Obesitas
Merah → Tidak Obesitas

Threshold: 0.5

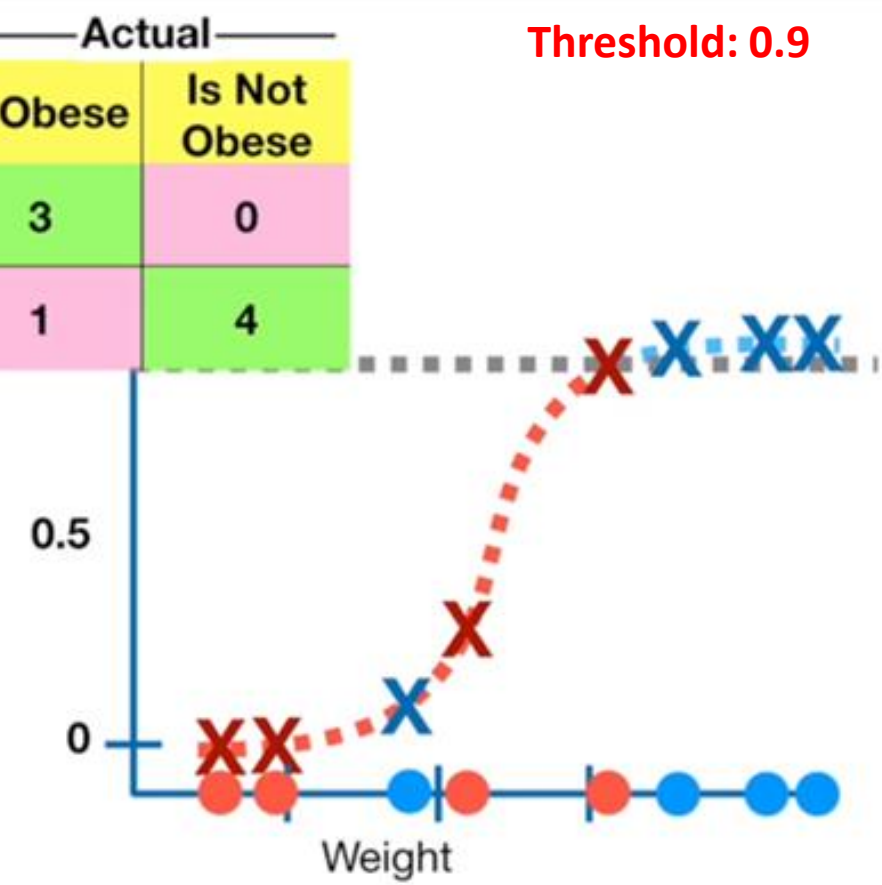
		Actual	
		Is Obese	Is Not Obese
Predicted	Is Obese	3	1
	Is Not Obese	1	3



ROC: Example

Kasus: Klasifikasi Tikus Obesitas/Tidak
Kelas: Biru → Obesitas
Merah → Tidak Obesitas

		Actual	
		Is Obese	Is Not Obese
Predicted	Is Obese	3	0
	Is Not Obese	1	4



ROC: Example

Kasus: Klasifikasi Tikus Obesitas/Tidak
Kelas: Biru → Obesitas
Merah → Tidak Obesitas

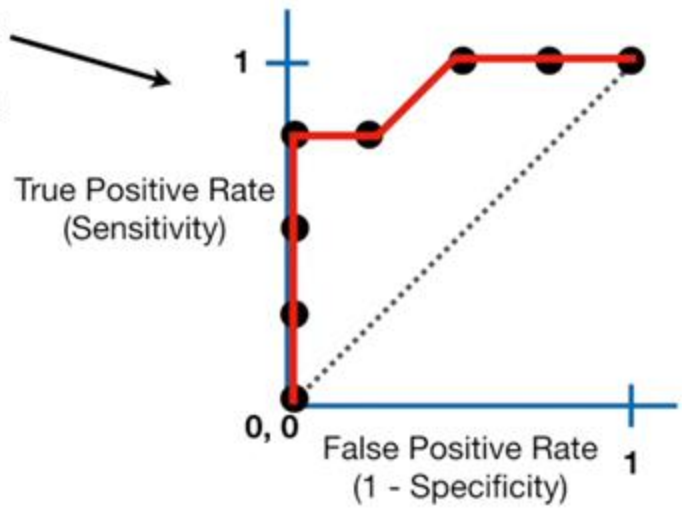
$$\text{TPR /Recall / Sensitivity} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{FPR} = 1 - \text{Specificity}$$

$$= \frac{FP}{TN + FP}$$

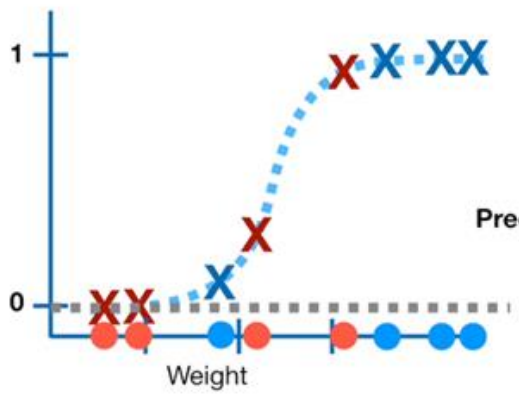
So instead of being overwhelmed with confusion matrices, **Receiver Operator Characteristic** (**ROC**) graphs provide a simple way to summarize all of the information.



Kita plot TPR & FPR untuk setiap kemungkinan Treshold mulai dari 0 s.d. 1

ROC: Example

Kasus: Klasifikasi Tikus Obesitas/Tidak
Kelas: Biru → Obesitas
Merah → Tidak Obesitas



		Actual: Is Obese	Actual: Is Not Obese
Predicted	Is Obese	4	4
	Is Not Obese	0	0

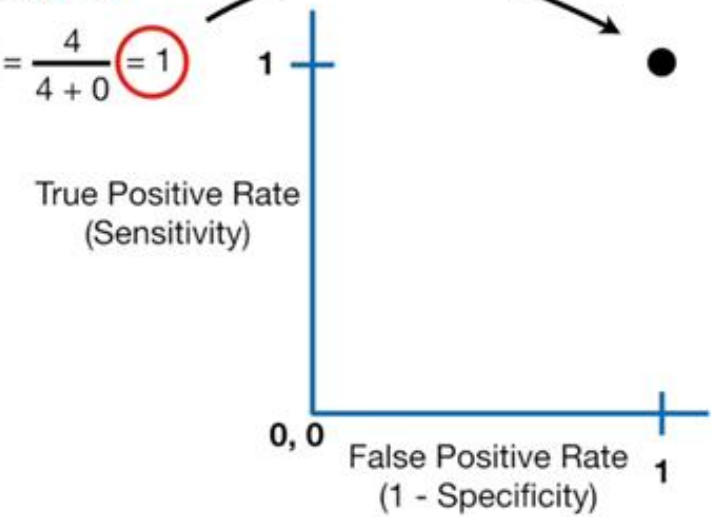
Kita plot TPR & FPR untuk setiap kemungkinan Treshold

Contoh utk Treshold = 0

True Positive Rate = Sensitivity = $\frac{4}{4+0} = 1$

False Positive Rate = 1 - Specificity = $\frac{4}{4+0} = 1$

Now plot a point at **1, 1.**



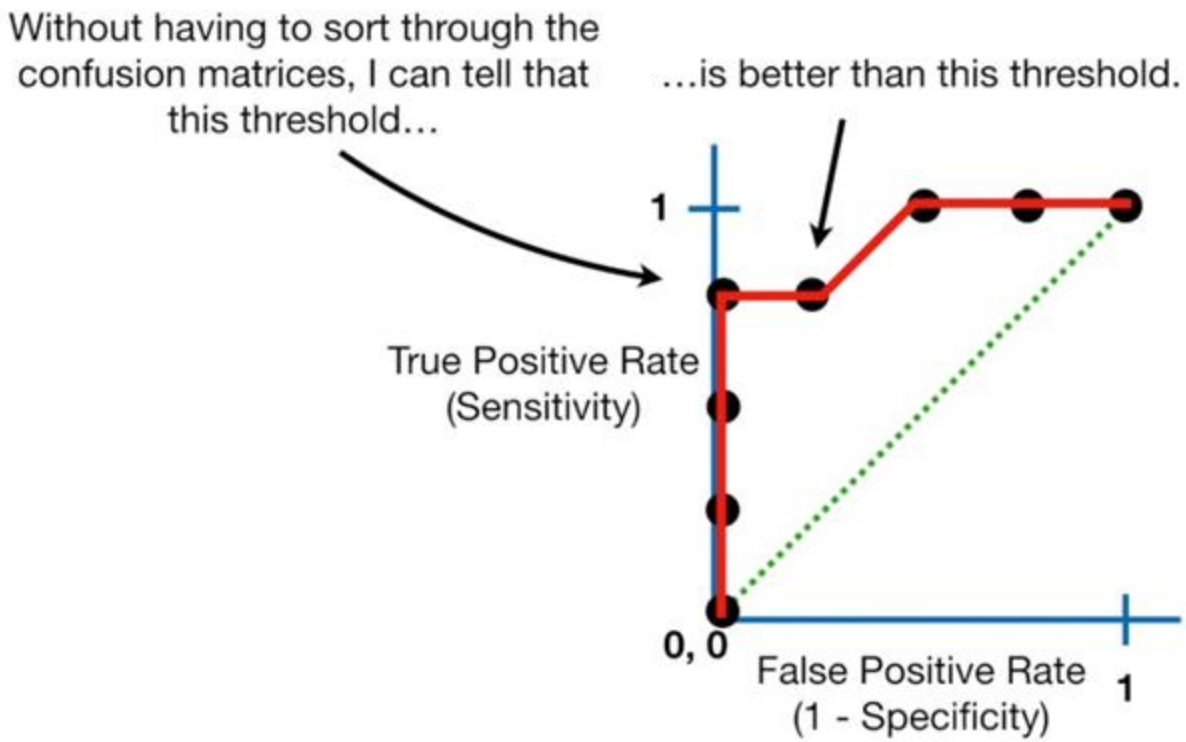
TPR /Recall / Sensitivity = $\frac{TP}{TP + FN}$

Specificity = $\frac{TN}{TN + FP}$

FPR = 1 - Specificity

$= \frac{FP}{TN + FP}$

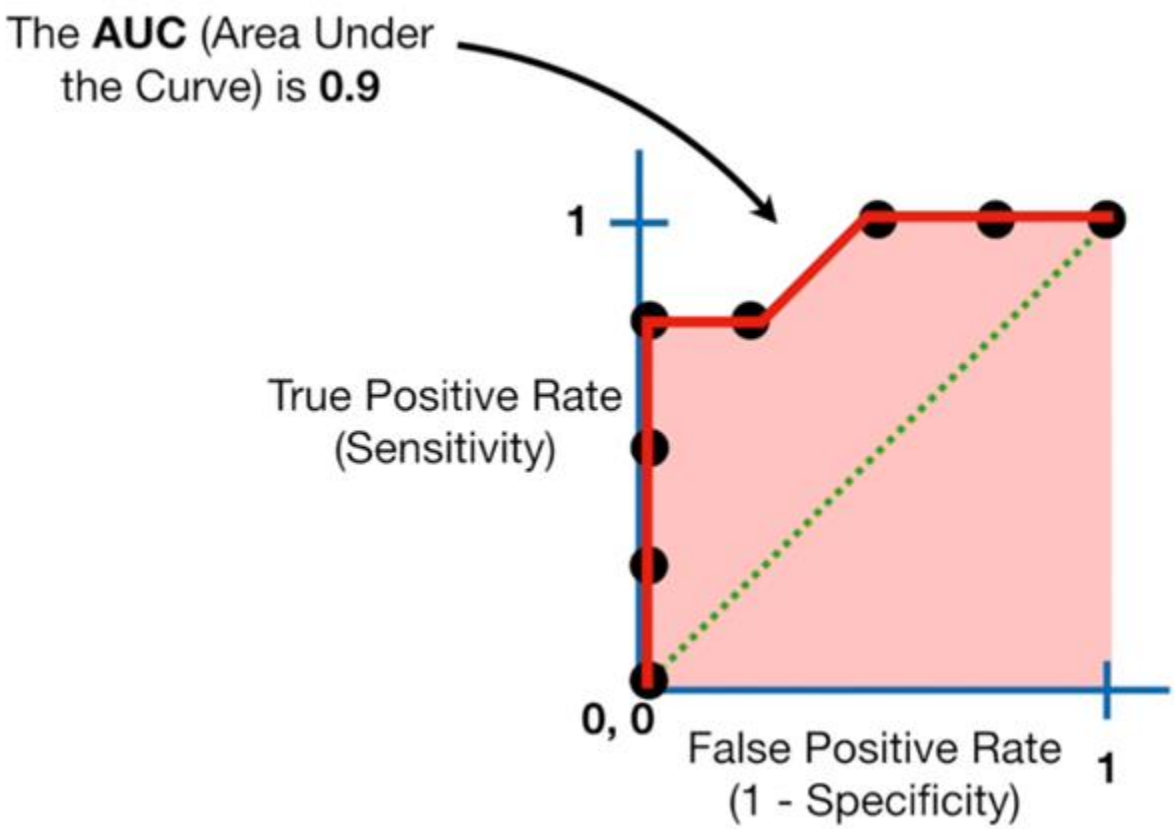
ROC: Example



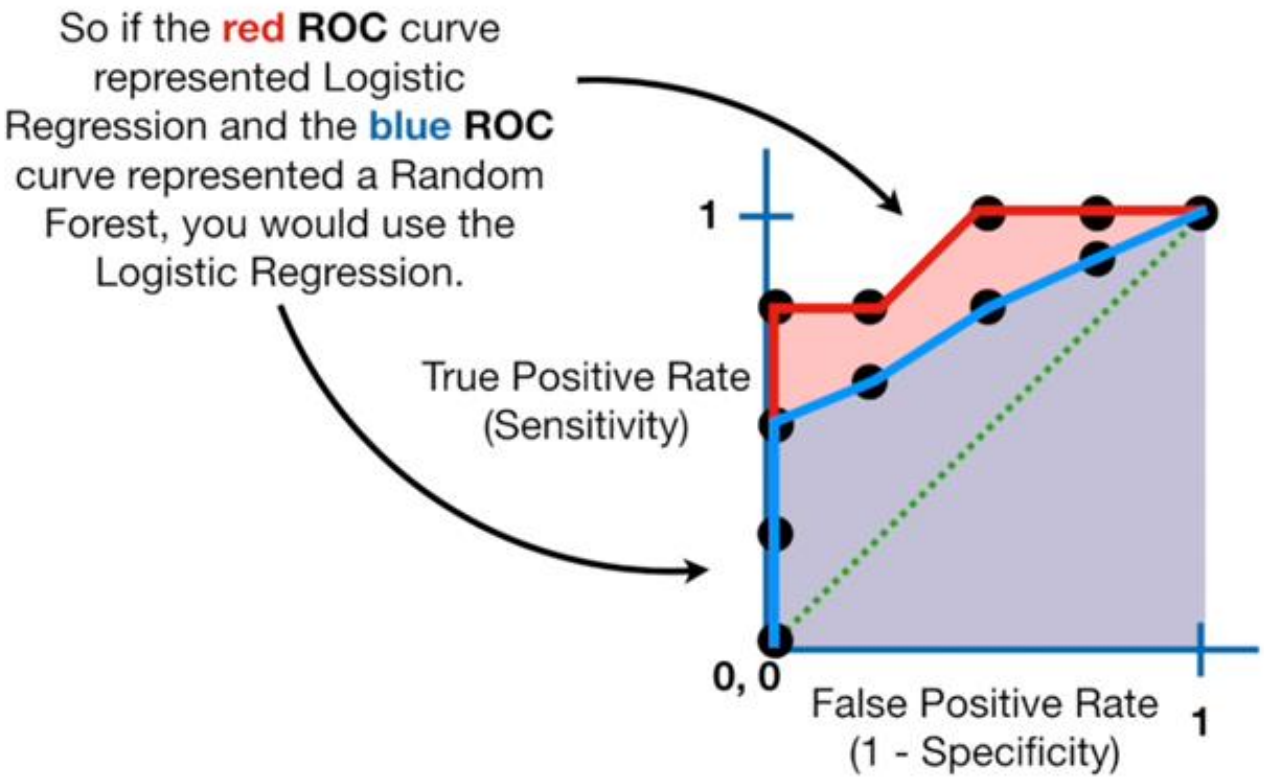
Semakin tinggi TPR dan semakin kecil FPR maka thresholdnya semakin bagus

AUC

- AUC: area under the ROC curve



Perbandingan Model dg AUC



Semakin besar AUC suatu model, maka performanya semakin baik, karena untuk setiap titik dia memiliki TP lebih tinggi dan atau FP lebih rendah

AUC: Justifikasi

- Terkadang untuk data yang imbalance, orang-orang lebih memilih menggunakan Precision daripada False Positive Rate.
- Precision tidak memiliki variable True Negatives dalam kalkulasinya.
  Precision = TP / (TP + FP)

Multiclass Confusion Matrix

		True/Actual		
		Cat ( )	Fish ( )	Hen ( )
Predicted	Cat ( )	4	6	3
	Fish ( )	1	2	0
	Hen ( )	1	2	6

Multiclass Prec-Rec

		True/Actual		
		Cat ( )	Fish ( )	Hen ( )
Predicted	Cat ( )	4	6	3
	Fish ( )	1	2	0
	Hen ( )	1	2	6

Prec = 4/(4+6+3)
= 4/13

Rec = 4/(4+1+1)
= 4/6
= 2/3

Multiclass Prec-Rec

		True/Actual		
		Cat ( )	Fish ( )	Hen ( )
Predicted	Cat ( )	4	6	3
	Fish ( )	1	2	0
	Hen ( )	1	2	6

		True/Actual		
		Cat ( )	Fish ( )	Hen ( )
Predicted	Cat ( )	4	6	3
	Fish ( )	1	2	0
	Hen ( )	1	2	6

Multiclass Prec-Rec

		True/Actual		
		Cat ( )	Fish ( )	Hen ( )
Predicted	Cat ( )	4	6	3
	Fish ( )	1	2	0
	Hen ( )	1	2	6

Class	Precision	Recall
Cat	30.8%	66.7%
Fish	66.7%	20.0%
Hen	66.7%	66.7%

Macro Prec-Rec

Class	Precision	Recall
Cat	30.8%	66.7%
Fish	66.7%	20.0%
Hen	66.7%	66.7%

- Macro-precision = (30.8% + 66.7% + 66.7%) / 3 = 54.7%
- Macro-recall = (66.7% + 20% + 66.7%) / 3 = 51.1%

Multiclass F1-Score

Class	Precision	Recall	F1-score
Cat	30.8%	66.7%	42.1%
Fish	66.7%	20.0%	30.8%
Hen	66.7%	66.7%	66.7%

- Macro-F1 = (42.1% + 30.8% + 66.7%) / 3 = 46.5%

Thank You